

Norway

SLEIPNER

Volve F

F-9

F-9

Plan: F9 Rev 0.13 061207 tangent from 750m

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-9	North Reference:	Grid
Wellbore:	F-9	Survey Calculation Method:	Minimum Curvature
Design:	F9 Rev 0.13 061207 tangent from 750m	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-9			
Well Position	+N/-S	-1.96 m	Northing:	6,478,561.57 m
	+E/-W	1.11 m	Easting:	435,051.13 m
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m
			Water Depth:	91.00 m

Wellbore	F-9				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MNETICREFERENCE	25.11.2007	-2.25	71.61	50,362

Design	F9 Rev 0.13 061207 tangent from 750m			
Audit Notes:				
Version:	Phase:	PLAN	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-1.96	1.11	88.84

Survey Tool Program	Date	31.05.2010		
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
145.90	254.90	F9 Rev 0.13 061207 tangent from 750m (F)	Keeper, stat	Gyro Tool from SDC
254.90	900.00	F9 Rev 0.13 061207 tangent from 750m (F)	Keeper, stat	Gyro Tool from SDC
900.00	1,114.25	F9 Rev 0.13 061207 tangent from 750m (F)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)

Planned Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
145.90	0.00	0.00	145.90	-1.96	1.11	0.00	0.000	0.000	0.000
254.90	0.00	0.00	254.90	-1.96	1.11	0.00	0.000	0.000	0.000
532.90	0.00	0.00	532.90	-1.96	1.11	0.00	0.000	0.000	0.000
600.90	4.00	90.00	600.84	-1.96	3.49	2.37	1.765	1.765	0.000
709.69	11.37	76.12	708.59	0.62	17.71	16.65	2.081	2.032	-3.828
750.00	11.37	76.12	748.11	2.53	25.43	24.40	0.000	0.000	0.000
883.51	11.37	76.12	879.00	8.84	50.98	50.07	0.000	0.000	0.000
1,074.25	11.37	76.12	1,066.00	17.86	87.49	86.75	0.000	0.000	0.000
1,114.25	11.37	76.12	1,105.22	19.75	95.14	94.45	0.000	0.000	0.000

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Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-9	North Reference:	Grid
Wellbore:	F-9	Survey Calculation Method:	Minimum Curvature
Design:	F9 Rev 0.13 061207 tangent from 750m	Database:	Production EDM P246N

Planned Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
Rev Utsira Top v2 07110 - plan misses target center by 48.98m at 888.77m MD (884.15 TVD, 9.09 N, 51.99 E) - Point	0.00	0.00	874.90	6.54	100.01	6,478,570.06	435,150.00	58° 26' 30.0716 N	1° 53' 21.0856 E
Utsira Top F9 - plan misses target center by 2.65m at 884.09m MD (879.57 TVD, 8.87 N, 51.09 E) - Point	0.00	0.00	879.90	6.48	50.00	6,478,570.00	435,100.00	58° 26' 30.0429 N	1° 53' 18.0031 E
Rev Utsira Top 071106 - plan misses target center by 2.59m at 884.10m MD (879.57 TVD, 8.87 N, 51.09 E) - Point	0.00	0.00	879.90	6.54	50.00	6,478,570.06	435,100.00	58° 26' 30.0449 N	1° 53' 18.0031 E
Utsira Base F9 - plan misses target center by 318.40m at 1114.25m MD (1105.21 TVD, 19.75 N, 95.14 E) - Point	0.00	0.00	1,058.90	1.52	409.62	6,478,565.04	435,459.50	58° 26' 30.0742 N	1° 53' 40.1716 E

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
221.70	221.70	30"	30.000	36.000	
905.85	900.90	13 3/8"	13.375	17.500	
1,109.85	1,100.90	9 5/8"	9.625	12.250	

Formations					
Measured Depth (m)	Vertical Depth (m)	Name	Lithology	Dip (°)	Dip Direction (°)
883.51	879.00	Top Utsira		0.00	
1,074.25	1,066.00	Base Utsira		0.00	

Checked By: _____	Approved By: _____	Date: _____
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